

Programming Assignment 1

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ERP ID: 26106

Course: Introduction to Blockchain

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So, this is basically a document with all the OUTPUT PROOFS. My code file is attached too.

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PROOF 1: test.py got 40/40 marks

```
python testCode.py
Enter your erp id: 26106
Connected to backend
9889 9889 5
870957153513604955099783510612415718372330343087099090278033937281925992370 0
25710847192128860711870250207729231661178476158772968986907201458313275253420 0
40189016485892926282548271016069919390207554076965828391660538936131874139403 0
43558234995138531315276814253345165558232202230712793313817517100022236863201 0
61368317325581018485444567640877575581714997222663672800401752329642593681898 0
68957154921550612234866922340053330019704169724720129011905449946148775763014 0
85128033328377724622986738793818727043753474774787377171671560282379194970854 0
87316322514699652727249331442542001328835646615455141794091667247614728216777 0
97900424532147900438859704322823053922997230128479610197495596665802299846947 5000000000
99652275076723255094766834832004149220812205664595801229873801477016512014599 0
870957153513604955099783510612415718372330343087099090278033937281925992370 5000000000
25710847192128860711870250207729231661178476158772968986907201458313275253420 5000000000
40189016485892926282548271016069919390207554076965828391660538936131874139403 5000000000
43558234995138531315276814253345165558232202230712793313817517100022236863201 5000000000
61368317325581018485444567640877575581714997222663672800401752329642593681898 5000000000
68957154921550612234866922340053330019704169724720129011905449946148775763014 5000000000
85128033328377724622986738793818727043753474774787377171671560282379194970854 5000000000
87316322514699652727249331442542001328835646615455141794091667247614728216777 5000000000
97900424532147900438859704322823053922997230128479610197495596665802299846947 5000000000
99652275076723255094766834832004149220812205664595801229873801477016512014599 5000000000
870957153513604955099783510612415718372330343087099090278033937281925992370 5000000000
25710847192128860711870250207729231661178476158772968986907201458313275253420 5000000000
40189016485892926282548271016069919390207554076965828391660538936131874139403 5000000000
43558234995138531315276814253345165558232202230712793313817517100022236863201 5000000000
```

```
61368317325581018485444567640877575581714997222663672800401752329642593681898 9999999835
68957154921550612234866922340053330019704169724720129011905449946148775763014 5000000000
85128033328377724622986738793818727043753474774787377171671560282379194970854 5000000000
87316322514699652727249331442542001328835646615455141794091667247614728216777 5000000000
97900424532147900438859704322823053922997230128479610197495596665802299846947 0
99652275076723255094766834832004149220812205664595801229873801477016512014599 5000000000
870957153513604955099783510612415718372330343087099090278033937281925992370 5000000000
25710847192128860711870250207729231661178476158772968986907201458313275253420 7969690844
40189016485892926282548271016069919390207554076965828391660538936131874139403 8573843577
43558234995138531315276814253345165558232202230712793313817517100022236863201 5000000000
61368317325581018485444567640877575581714997222663672800401752329642593681898 11524896630
68957154921550612234866922340053330019704169724720129011905449946148775763014 6426155594
85128033328377724622986738793818727043753474774787377171671560282379194970854 0
87316322514699652727249331442542001328835646615455141794091667247614728216777 0
97900424532147900438859704322823053922997230128479610197495596665802299846947 0
99652275076723255094766834832004149220812205664595801229873801477016512014599 5505410971
Bad transaction was trying to: 43558234995138531315276814253345165558232202230712793313817517100022236863201
870957153513604955099783510612415718372330343087099090278033937281925992370 1904043603
25710847192128860711870250207729231661178476158772968986907201458313275253420 2531522792
40189016485892926282548271016069919390207554076965828391660538936131874139403 9942931921
43558234995138531315276814253345165558232202230712793313817517100022236863201 1792825777
61368317325581018485444567640877575581714997222663672800401752329642593681898 3987997543
68957154921550612234866922340053330019704169724720129011905449946148775763014 60512845
85128033328377724622986738793818727043753474774787377171671560282379194970854 13396149551
87316322514699652727249331442542001328835646615455141794091667247614728216777 4671489268
97900424532147900438859704322823053922997230128479610197495596665802299846947 47783348
99652275076723255094766834832004149220812205664595801229873801477016512014599 11664420824
```

Your Marks: 40 out of 40

> python testCode.py

Enter your erp id: 26106

Connected to backend

9889 9889 5

870957153513604955099783510612415718372330343087099090278033937281925992370 0

25710847192128860711870250207729231661178476158772968986907201458313275253420 0

40189016485892926282548271016069919390207554076965828391660538936131874139403 0

43558234995138531315276814253345165558232202230712793313817517100022236863201 0

61368317325581018485444567640877575581714997222663672800401752329642593681898 0

68957154921550612234866922340053330019704169724720129011905449946148775763014 0

85128033328377724622986738793818727043753474774787377171671560282379194970854 0

87316322514699652727249331442542001328835646615455141794091667247614728216777 0

97900424532147900438859704322823053922997230128479610197495596665802299846947 5000000000

99652275076723255094766834832004149220812205664595801229873801477016512014599 0
870957153513604955099783510612415718372330343087099090278033937281925992370 5000000000
25710847192128860711870250207729231661178476158772968986907201458313275253420 5000000000
40189016485892926282548271016069919390207554076965828391660538936131874139403 5000000000
43558234995138531315276814253345165558232202230712793313817517100022236863201 5000000000
61368317325581018485444567640877575581714997222663672800401752329642593681898 5000000000
68957154921550612234866922340053330019704169724720129011905449946148775763014 5000000000
85128033328377724622986738793818727043753474774787377171671560282379194970854 5000000000
87316322514699652727249331442542001328835646615455141794091667247614728216777 5000000000
97900424532147900438859704322823053922997230128479610197495596665802299846947 5000000000
99652275076723255094766834832004149220812205664595801229873801477016512014599 5000000000
870957153513604955099783510612415718372330343087099090278033937281925992370 5000000000
25710847192128860711870250207729231661178476158772968986907201458313275253420 5000000000
40189016485892926282548271016069919390207554076965828391660538936131874139403 5000000000
43558234995138531315276814253345165558232202230712793313817517100022236863201 5000000000
61368317325581018485444567640877575581714997222663672800401752329642593681898 9999999835
68957154921550612234866922340053330019704169724720129011905449946148775763014 5000000000
85128033328377724622986738793818727043753474774787377171671560282379194970854 5000000000
87316322514699652727249331442542001328835646615455141794091667247614728216777 5000000000
97900424532147900438859704322823053922997230128479610197495596665802299846947 0
99652275076723255094766834832004149220812205664595801229873801477016512014599 5000000000
870957153513604955099783510612415718372330343087099090278033937281925992370 5000000000
25710847192128860711870250207729231661178476158772968986907201458313275253420 7969690844
40189016485892926282548271016069919390207554076965828391660538936131874139403 8573843577
43558234995138531315276814253345165558232202230712793313817517100022236863201 5000000000
61368317325581018485444567640877575581714997222663672800401752329642593681898 11524896630
68957154921550612234866922340053330019704169724720129011905449946148775763014 6426155594
85128033328377724622986738793818727043753474774787377171671560282379194970854 0
87316322514699652727249331442542001328835646615455141794091667247614728216777 0
97900424532147900438859704322823053922997230128479610197495596665802299846947 0

99652275076723255094766834832004149220812205664595801229873801477016512014599 5505410971

Bad transaction was trying to:

43558234995138531315276814253345165558232202230712793313817517100022236863201

870957153513604955099783510612415718372330343087099090278033937281925992370 1904043603

25710847192128860711870250207729231661178476158772968986907201458313275253420 2531522792

40189016485892926282548271016069919390207554076965828391660538936131874139403 9942931921

43558234995138531315276814253345165558232202230712793313817517100022236863201 1792825777

61368317325581018485444567640877575581714997222663672800401752329642593681898 3987997543

68957154921550612234866922340053330019704169724720129011905449946148775763014 60512845

85128033328377724622986738793818727043753474774787377171671560282379194970854 13396149551

87316322514699652727249331442542001328835646615455141794091667247614728216777 4671489268

97900424532147900438859704322823053922997230128479610197495596665802299846947 47783348

99652275076723255094766834832004149220812205664595801229873801477016512014599 11664420824

Your Marks: 40 out of 40

Then main.py was run, “rl” to fetch the latest chain of 696 blocks, and “validate” was run to validate the file, and “print” was done to print the chain. Then mining began.

PROOF 2: Mining was successful

> mine

Mining... this may take a while with difficulty 4

Still mining... tried up to nonce 50000

Still mining... tried up to nonce 100000

Still mining... tried up to nonce 150000

Still mining... tried up to nonce 200000

Still mining... tried up to nonce 250000

Still mining... tried up to nonce 300000

Still mining... tried up to nonce 350000

Still mining... tried up to nonce 400000

Still mining... tried up to nonce 450000
Still mining... tried up to nonce 500000
Still mining... tried up to nonce 550000
Still mining... tried up to nonce 600000
Still mining... tried up to nonce 650000
Still mining... tried up to nonce 700000
Still mining... tried up to nonce 750000
Still mining... tried up to nonce 800000
Still mining... tried up to nonce 850000
Still mining... tried up to nonce 900000
Still mining... tried up to nonce 950000
Still mining... tried up to nonce 1000000
Still mining... tried up to nonce 1050000
Still mining... tried up to nonce 1100000
Still mining... tried up to nonce 1150000
Still mining... tried up to nonce 1200000
Still mining... tried up to nonce 1250000
Still mining... tried up to nonce 1300000

block hash found

0000c88055d80223236e80abbffc460a381b0a5e0d492ca30f722e9ccf74b1b3

Block successfully mined!

PROOF 3: chain successfully sent to backend

> send state

State sent to backend

PROOF 4: chain fetched and printed to see if my block was added

> rl

Requested longest chain, check pending_chains folder

Backend has no longer chain yet, please try again after some time...

> validate

No pending chains found to validate.

> print

... excluded al 686 blocks

Block index # 687

Transaction number 1276 with hash

106897296527924764199667125552980890037746170738505784713432458442547679971572

Transaction number 1279 with hash

4348696049650101035633769778528607889714584734871062526600404328859556934820

Transaction number 1283 with hash

15206552066005789095658745641384009236848085361028234329244152637804893412115

Transaction number 1290 with hash

98690495398309002924443911061725567558432810741273202135647905795356076794112

Transaction number 1295 with hash

46911183463424891503636513496414637039836375655325814063081513851381921458587

nonce: 1300460

previous_hash: 00004a3ec2f86ddc5b0bbe0573d23906e9adbef9d12a552d27b64472b145469c

hash: 0000c88055d80223236e80abbffc460a381b0a5e0d492ca30f722e9ccf74b1b3

Miner: 26106

PROOF 5: checked chain for my ERP using commands

> python -c "import glob,os,pickle; from FullNode import FullNode;

files=sorted(glob.glob('valid_chain/block*.block'), key=lambda x:int(os.path.basename(x)[5:-6])); f=[f for f in files

```
if os.path.basename(f)=='block687.block'][0]; b=pickle.load(open(f,'rb')); n=FullNode('26106'); print('index=',  
b.index); print('miner=', b.miner); print('hash=', n.computeBlockHash(b))"
```

```
index= 687
```

```
miner= 26106
```

```
hash= 0000c88055d80223236e80abbffc460a381b0a5e0d492ca30f722e9ccf74b1b3
```

```
> python -c "import glob,os,pickle; files=sorted(glob.glob('valid_chain/block*.block'), key=lambda  
x:int(os.path.basename(x)[5:-6])); found=[]; [found.append((b.index,b.miner)) for f in files for b in  
[pickle.load(open(f,'rb'))] if str(b.miner)=='26106']; print(found[-20:]); print('total matches =', len(found))"
```

```
[(687, '26106')]
```

```
total matches = 1
```